Sensors

Sensor: Provides a signal (electrical, optical) depending on the property (physical, chemical, biological) to be measured.

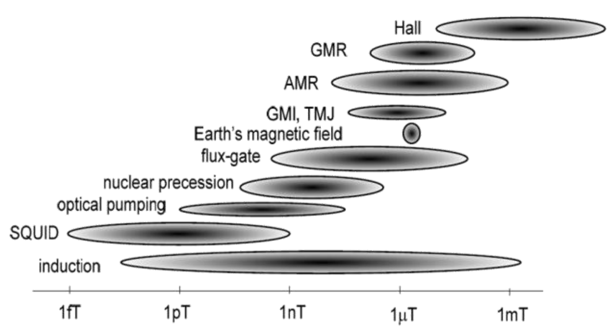
Types:
- -Magnetic field sensors
- -Distance and position sensors
- -Acceleration sensors
- -Temperature sensors
- -Light sensors
- -Pressure sensors
- -Force sensors
- -Elongation sensors
- -Humidity sensors
- -Chemical, biological sensors

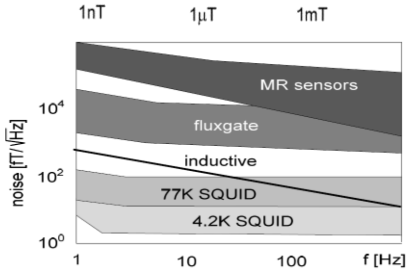
Features: sensitivity, measuring range, selectivity, non-reactivity, stability.

All around us: Smart devices, Internet of Things.

Magnetic field sensors

- Inductive sensors
- Magnetoresistive sensors: AMR, GMR, TMR
- Hall sensors
- SQUID
- Nitrogen vacancy centers





Typical magnetic field ranges:

- 100fT – 1 pT: human brain's magnetic field
- 30-60uT geomagnetic field
- 10mT: field of a fridge magnet
- 1T: magnetic field in the transformer core
- 1.25T: strong rare earth magnets
- 14T: Field in a superconducting solenoid magnet (e.g. MRI)

Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

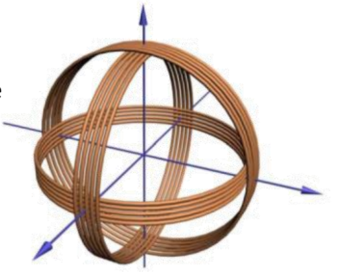
Magnetic field sensors

Factor (tesla)	SI prefix	Value (SI units)	Value (cgs units)	Item
10^{-18}	attotesla	5 aT	50 fG	SQUID magnetometers on Gravity Probe B gyroscopes measure fields at this level over several days of averaged measurements[2]
10^{-15}	femtotesla	2 fT	20 pG	SQUID magnetometers on Gravity Probe B gyros measure fields at this level in about one second
10^{-12}	picotesla	100 fT to 1 pT	1 nG to 10 nG	human brain magnetic field
10^{-11}		10 pT	100 nG	In September 2006, NASA found "potholes" in the magnetic field in the heliosheath around our solar system that are 10 picoteslas as reported by Voyager 1[3]
10^{-9}	nanotesla	100 pT to 10 nT	1 μG to 100 μG	magnetic field strength in the heliosphere
10^{-6}	microtesla	24 μT	240 mG	strength of magnetic tape near tape head
10^{-5}		31 μT	310 mG	strength of Earth's magnetic field at 0° latitude (on the equator)
		58 μT	580 mG	strength of Earth's magnetic field at 50° latitude
10^{-3}	millitesla	0.5 mT	5 G	the suggested exposure limit for cardiac pacemakers by American Conference of Governmental Industrial Hygienists (ACGIH)
		5 mT	50 G	the strength of a typical refrigerator magnet [4]
10^{-1}		0.15 T	1.5 kG	the magnetic field strength of a sunspot
10^{0}	tesla	1 T to 2.4 T	10 kG to 24 kG	coil gap of a typical loudspeaker magnet.[5]
		1 T to 2 T	10 kG to 20 kG	inside the core of a modern 60 Hz power transformer[6][7]
		1.25 T	12.5 kG	strength of a modern neodymium–iron–boron ($Nd_2Fe_{14}B$) rare earth magnet. A coin-sized neodymium magnet can lift more than 9 kg, pinch skin and erase credit cards.[8]
		1.5 T to 3 T	15 kG to 30 kG	strength of medical magnetic resonance imaging systems in practice, experimentally up to 8 T[9][10]
		9.4 T	94 kG	Modern high resolution research magnetic resonance imaging system
10^{1}		11.7 T	117 kG	field strength of a 500 MHz NMR spectrometer
		16 T	160 kG	strength used to levitate a frog[11]
		23.5 T	235 kG	field strength of a 1 GHz NMR spectrometer[12]
		36.2 T	362 kG	strongest continuous magnetic field produced by non-superconductive resistive magnet.[13]
		45 T	450 kG	strongest continuous magnetic field yet produced in a laboratory (Florida State University's National High Magnetic Field Laboratory in Tallahassee, USA).[14]
10^{2}		129.053 T	1.29 MG	strongest (pulsed) magnetic field yet obtained non-destructively in a laboratory (Keshe Foundation, SSI, with less than 20 watts of power and 10ml or less of gans nano fluid , ITALY)[15]
		730 T	7.3 MG	strongest pulsed magnetic field yet obtained in a laboratory, destroying the used equipment, but not the laboratory itself (Institute for Solid State Physics, Tokyo)
10^{3}	kilotesla	2.8 kT	28 MG	strongest (pulsed) magnetic field ever obtained (with explosives) in a laboratory (VNIIEF in Sarov, Russia, 1998)[16]

Magnetic field sensors: inductive sensors

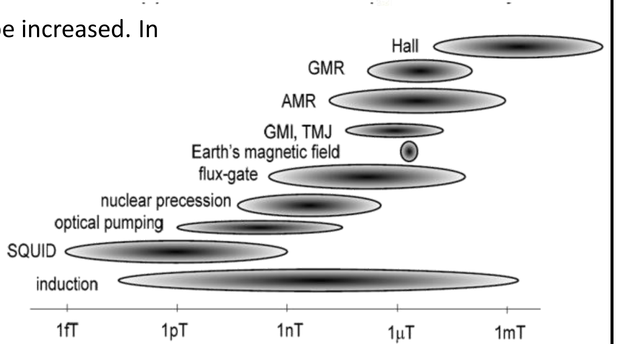
Principle: a variable B field in a coil causes an induced voltage.

$B = B_0 \sin \omega\, t$ oscillating field yields an induced voltage amplitude: $V_0 = -nA\omega B_0$, where N is the number of turns of the coil and A is the cross-section.



Advantages (+) and disadvantages (-):

+ Broad application range, simple, the 3 field directions are easily detectable
- To increase sensitivity, size must be increased. In geophysics meter-sized inductive sensors are also used.
- Detection requires changing magnetic field.
  → to measure static field the sensor must be moved

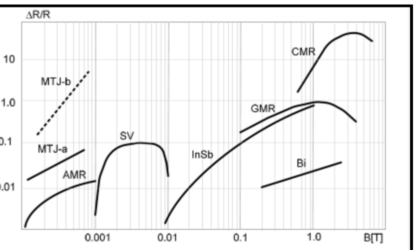


Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

Magnetoresistive magnetic field sensors

Magnetic field sensors based on resistance change.
They can rely on several physical phenomena:

- **AMR:** **A**nisotropic **M**agneto**r**esistance
- **GMR:** Giant Magnetoresistance, **SV:** **S**pin **V**alve
- **MTJ:** **M**agnetic **T**unnel **J**unction
- **CMR**: **C**olossal **M**agneto**r**esistance

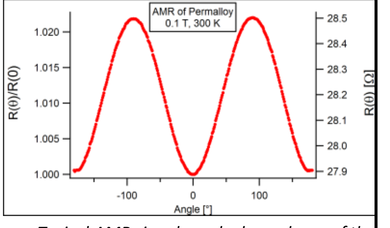


Various physical phenomena associated with magnetic-field-induced resistance-change. Relative resistance changes are shown as a function of magnetic field covering various field ranges.

AMR, Anisotropic **M**agneto**r**esistance
1857 Kelvin: The resistance of a ferromagnet (F) is different if the current direction is parallel or perpendicular to the direction of magnetization.
Reason: It is due to magnetism and so-called spin-orbit interaction. For example, the electron scatters more strongly when it moves in the direction of M. In general, the resistance is higher in the direction of M.
Typical angle dependence of the AMR phenomenon:

$$\rho(\theta)= \rho_{perp} + (\rho_{paral} - \rho_{perp}) \cos^2(\theta) ,$$

where θ is the angle between the **M** magnetization and the **j** current density. The ρ resistivity is ρ_{perp} or ρ_{paral} if the current is perpendicular or parallel to **M**.

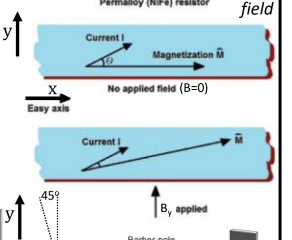


Typical AMR signal: angle dependence of the resistance in Permalloy (20% Fe, 80% Ni)

Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

Anisotropic Magnetoresistance (AMR) magnetic field sensors

Magnetization and current direction in a ferromagnetic (F) tape without and in the presence of an external magnetic field



In the case of a thin narrow but long ferromagnetic layer, the shape anisotropy determines the preferred magnetization direction. For example, magnetization sets in the longitudinal direction (see the easy axis in the figure).

Principle: an external field is applied perpendicular to the internal magnetization direction. This field somewhat modifies the magnetization direction, i.e. the angle of the current and **M,** and the related magnetoresistance changes.

Problem: naturally the current would also flow in the longitudinal direction, yielding θ=0. At this working point ρ shows a quadratic θ dependence (see previous slide) instead of a linear relation. The best linear relation is obtained at θ=45°.

Solution: Barber-pole geometry. We evaporate well conducting metallic stripes on the top of the magnetic layer with 45° orientation. These stripes are much better conductors than the magnetic layer underneath, i.e. the current wishes to minimize its route between two neighbor metallic stripes, i.e. it flows at θ= 45° direction in the magnetic layer in those segments, where no metal is on the top. Otherwise, it flows in the metallic stripe instead of the magnetic layer.

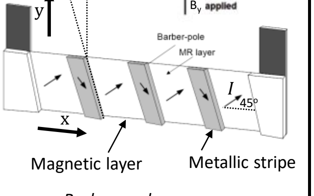
Problem 2: the resistance of the AMR sensor also depends on the temperature.

Practical implementation: Wheatstone – bridge with Barber poles of 45o and -45o orientation. V+ is applied, and V_{OUT} is measured. In this arrangement, the resistance of the 4 AMR sensors will change similarly with the external temperature change.

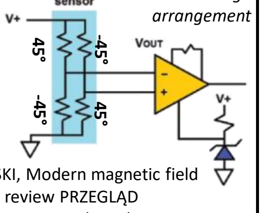
Advantages, disadvantages:

+ Simple, long-known principle, cheap.
+ Better sensitivity than GMR sensors. E.g. Phillips KMZ10B AMR sensor has 16μV/μT sensitivity at V+=5V supply voltage.

- Small resistance change, <2%
- Sensitive to longitudinal field components. Operational condition: $B_y < 0.1B_x$
- In large B field the sensor can be demagnetized → malfunction



Barber –pole

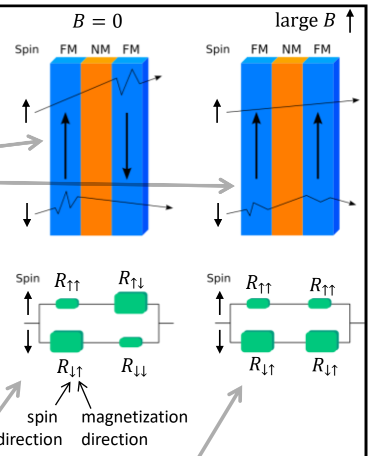


Wheatstone bridge arrangement

S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

Giant Magnetoresistance (GMR) magnetic field sensors

Source: wikipedia



Nobel prize 2007: Albert Fert & Peter Grünberg

Arrangement: Ferromagnetic (FM) thin layers separated by nonmagnetic (NM) metallic layer. In B=0 field the FM layers align antiferromagnetically (neighbor FM layers have opposite, antiparallel, AP magnetization). In large field all the FM layers align with the field direction (parallel, P magnetization.

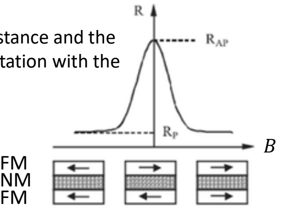
Operation: if the magnetization is opposite to the electron spin, the electrons scatter a lot in the F layer, yielding a large resistance ($R_{\uparrow\downarrow} = R_{\downarrow\uparrow}$). The same magnetization and spin orientation yields a smaller resistance due to the less scattering ($R_{\uparrow\uparrow} = R_{\downarrow\downarrow} < R_{\uparrow\downarrow}$). The phenomenon is similar to optical polarizing filters: little light passes through crossed polarizing filters, while much light passes through parallel ones. Here the ↑↓ spin and magnetization orientation filters the electrons. It can be seen in a simple resistor model, that in large field the resistance is smaller (the structure is more transparent for at least one of the spin orientations, whereas in zero field both spin orientations are filtered in one of the layers).

$$R_{\mathrm{B=0}} = R_{AP} = \frac{R_{\uparrow\downarrow} + R_{\uparrow\uparrow}}{2} \qquad R_{\mathrm{large\ B}} = R_P = \frac{2R_{\uparrow\downarrow} R_{\uparrow\uparrow}}{R_{\uparrow\downarrow} + R_{\uparrow\uparrow}}$$

$$R_{\uparrow\uparrow} = R_{\downarrow\downarrow} = \bar{R}(1-\beta), \qquad R_{\uparrow\downarrow} = R_{\downarrow\uparrow} = \bar{R}(1+\beta)$$

$$R_{\mathrm{B=0}} = \bar{R} \qquad R_{\mathrm{large\ B}} = \bar{R}(1-\beta^2) < R_{\mathrm{B=0}}$$

Variation of the resistance and the magnetization orientation with the magnetic field

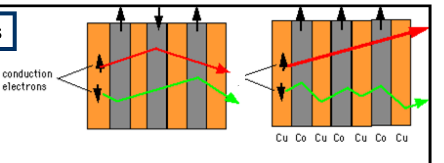


Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

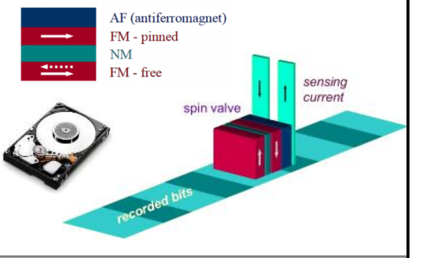
Original GMR, Spin valve and tunneling magnetoresistance sensors

Original GMR arrangement: multiple ferromagnetic layers used with a very thin nonmagnetic layer between them so that an antiferromagnetic coupling is established between the neighbor layers. A large field is required to align the magnetizations, i.e. the sensor is not very sensitive.

The ΔR resistance change can reach 50%.



Spin valve (SV) geometry: This is more common nowadays. There is a freely rotating FM layer, and another FM layer with magnetization pinned by a neighbor layer. In this arrangement the NM spacer layers should not be so thin as in the original geometry, as the magnetization alignment is not determined by this. The SV arrangement is used in the read heads of hard disk drives (HDD) and also in magnetic memories (MRAM).



Advantages and disadvantages of the SV geometry:

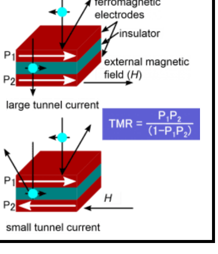
+ Small, more sensitive than the original GMR geomery, small energy consumption, fast response, weak temperature dependence
- More expensive

Tunneling Magnetoresistance (TMR) or Magnetic Tunnel Junction (MTJ) sensors:

Thin insulator between the two FM layers.

+ ΔR/R can reach 1000% in MgO tunnel junctions. Sensitive already in the mT field range.

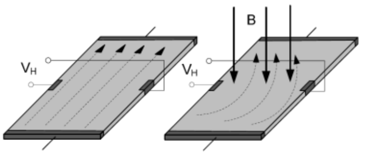
- Expensive heterostructure, small B range, large energy consumption



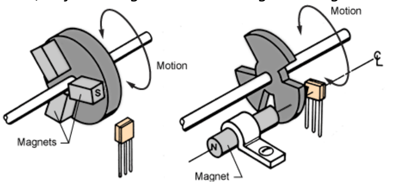
Sources: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)
https://www.electricalelibrary.com/en/2017/08/02/what-is-spintronics/

Hall sensors

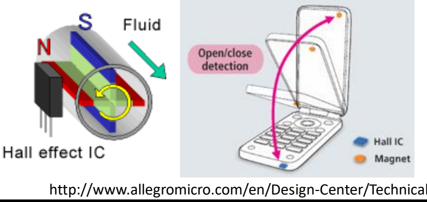
Hall voltage due to the Lorentz force in B field



Hall sensors as revolution counters. Magnets on the axle, or fixed magnet with rotating shielding disc



Flow meter (left) and openness sensor (right)



1879 Hall effect (see the slides on Quantized Hall effect for more details)

Principle: in B field the Lorentz force yields a perpendicular voltage drop on the sample (Hall voltage, V_H). The related Hall resistance in a 2D electron gas (see Metrology slides):

$$R_H = \frac{V_H}{I} = \frac{E_y}{j_x} = \frac{B}{en_{2D}}$$

Small n electron density is required for a large signal → semiconducting samples should be used.

Advantages/disadvantages:

+ Simple and cheap, the marked is dominated by Hall sensors
- Not too sensitive, <5mV/mT (but 50mV/mT is also achievable)
- Very strong temperature dependence due to the semiconducting nature of the sample
+ Size can be smaller than 1μm,
+ Small energy consumption
+ Non-invasive measurement, i.e. no ferromagnetic part which generates some magnetic field. Also lacks the demagnetization problem of the sensors with FM layers.
+ Si-based sensors are easily embedded in integrated circuits with amplifiers, stabilizers

Typical applications:

- Revolution counters e.g. in cars.
- Flow meters
- Openness sensors for mobile phones (very low power consumption <1.4μA)

http://www.allegromicro.com/en/Design-Center/Technical-Documents/Hall-Effect-Sensor-IC-Publications/Hall-Effect-IC-Application-Guide.aspx

Comparison of various magnetoresistive sensors

Typical price: <1USD/piece

- **Hall sensor**

- Small voltage signal and small sensitivity, strong temperature dependence

+ Cheap, small, low consumption

- **GMR and SV**

+ Small, sensitive, weak temperature dependence, low consumption, fast

- More expensive to produce
- Hard to measure field polarity, not very good for a compass

- **TMR**

+ Sensitive, small noise

- Narrow B field range, larger energy consumption, expensive

- **AMR**

2-3% R variation, Wheatstone bridge arrangement, mV output

+ high accuracy, weak temperature dependence

- Small signal, but it can be amplified. Large size.

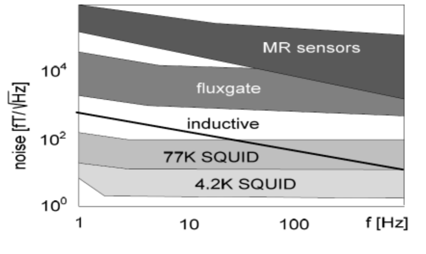
A Further compact, but non-magnetoresistive sensor:

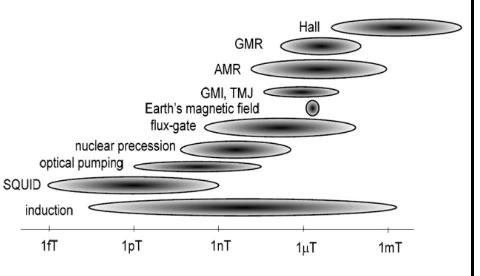
- **Lorentz force meter** (new Microelectromechanical System, MEMS sensor)

Detection of a small magnet's displacements due to magnetic field

+ CMOS compatible, easily integrated

- Resolution is limited by the noise of the electronics

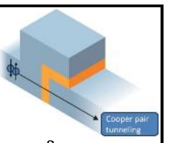




Y. Cai et al. Magnetometer basics for mobile phone applications, Sensors & Transducers

Superconducting Quantum Interference Device (SQUID)

see the Josephson effect on the metrology slides

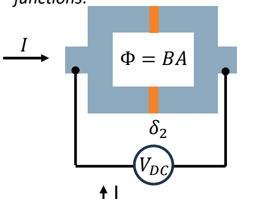


Principle: two parallel Josephson junctions (JJ). In zero field the phase difference would be the same δ_0 at the two JJs. A finite field introduces a different phase difference at the two arms: $\delta_1 = \delta_0 + \frac{2\pi e}{h}\Phi$, $\delta_2 = \delta_0 - \frac{2\pi e}{h}\Phi$, where e is the electron charge, h is Planck's constant, and $\Phi = BA$ is the enclosed flux with B being the magnetic field and A the area enclosed by the SQUID. Like this the total current of the SQUID is written as:

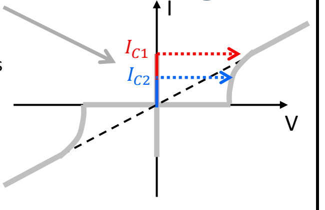
$$I = I_1 + I_2 = I_0[\sin(\delta_0 + 2\pi e\Phi/h) + \sin(\delta_0 - 2\pi e\Phi/h)] = 2I_0\cos(2\pi e\Phi/h)\sin\delta_0$$

This means, that the critical current of the SQUID is modulated by the enclosed flux according to the $\cos(2\pi e\Phi/h)$ function.

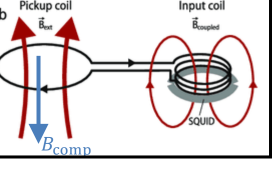
DC SQUID geometry with two Josephson junctions:



If we current bias the squid and increase the current, zero voltage drops on the SQUID as long as the critical current is not reached. Reaching the critical current, the SQUID turns to a non-superconducting state, where already a finite voltage drop is observed. As the magnetic field is varied, this crossover happens at different currents, and thus different voltages. This is periodic in the flux: taking A=1cm^2 device area one period corresponds to ΔB=2 10^{-11} T change in the magnetic field. This is an extremely sensitive device for measuring very small, femto-tesla field variations!



Usually, a loop of a closed superconducting wire is applied as a flux transformer. Due to its zero resistance the superconducting wire will preserve the overall enclosed flux (see demo experiment in the first lecture). The flux detected by the SQUID at the input coil equals the flux enclosed by the pickup coil divided by the number of turns at the input coil. Usually a feedback is also used: a B_{comp} compensating field keeps the flux zero at the pickup, and accordingly also at the SQUID. If the SQUID detects field variation, the compensating field is regulated with a PI loop to keep the flux zero.



Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)
IFCN-endorsed practical guidelines for clinical magnetoencephalography, DOI:10.1016/j.clinph.2018.03.042

Superconducting Quantum Interference Device (SQUID) - applications

- Extremely sensitive, fT fields can be detected!
- Important application area: investigation of brain functions

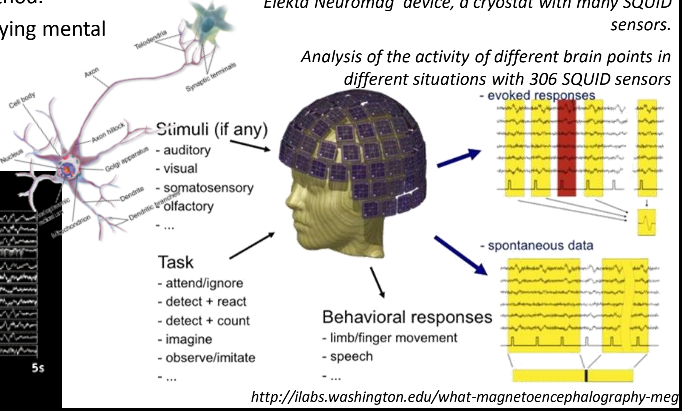
MEG (**Magnetoencephalography**)

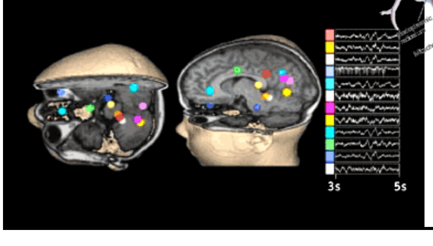
Advantages of MEG in brain research: both position and time of brain activity can be tracked in situ with ~mm spatial and ~sub-ms temporal resolution. It is a non-invasive method.

Brain functions can be studied, identifying mental diseases like epilepsy, stroke, etc. The process of learning can also be studied.

Elekta Neuromag device, a cryostat with many SQUID sensors.

Analysis of the activity of different brain points in different situations with 306 SQUID sensors

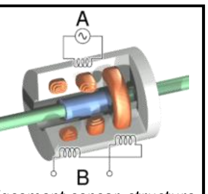




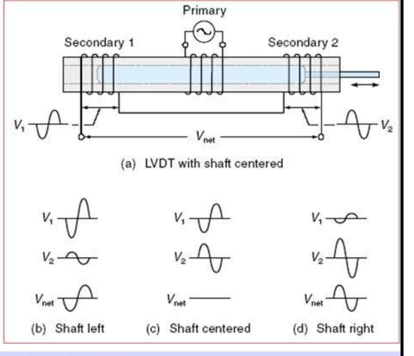
http://ilabs.washington.edu/what-magnetoencephalography-meg

Distance and displacement sensors

http://en.wikipedia.org/wiki/Linear_variable_differential_transformer



(Above) Inductive displacement sensor structure
(Below) Signal of the two pick-up coils at different positions of the iron core



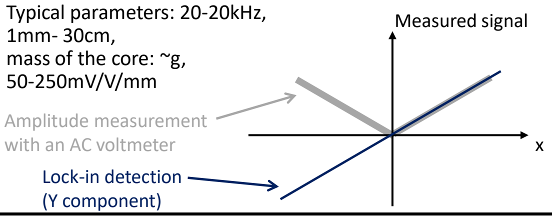
Types:

- Inductive, - Capacitive, - Laser light, - Ultrasound, - Radar, - Lidar

Inductive displacement sensor

(LVDT) Linear variable differential transformer

- Configuration: the ac signal applied to the driving coil (A) is detected by placing two oppositely wound coils (B) to the two sides of (A). These "pick-up" coils are connected in series. In the center of the coils there is an iron core, whose position is changed by the object to be measured by means of a rod.
- Principle: Mutual induction varies with the position of the iron core. If the iron core is right in the middle, the pick-up coils are balanced (shaft centered), no signal is detected. If the shaft moves to the left or to the right, a finite signal is detected, whose amplitude linearly increases with the displacement. If just the amplitude is detected, there is no difference between the left and right position, but by lock-in detection (e.g. measuring the Y component with respect to the driving) one obtains a true linear characteristics around zero displacement:
- Typical parameters: 20-20kHz,
  1mm- 30cm,
  mass of the core: ~g,
  50-250mV/V/mm

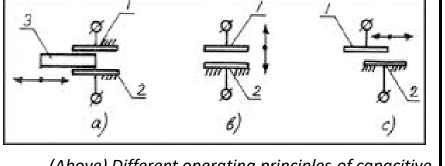


Capacitive displacement sensors

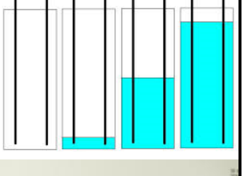
Principle: Capacity (C) depends on the distance (d) between the electrodes, surface area (A) and the dielectric constant of the intermediate region (ε_r):

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

Detection may rely on the variation of all these quantities (d, A and ε_r), see figures.



(Above) Different operating principles of capacitive displacement sensors: (a) a dielectric piece is actuated between the electrodes, (b) the electrode distance is varied, (c) the electrodes are moved parallel to each other
(Below) Liquid level measurement: the dielectric constant is different in liquid and in air



Application:

-In semiconductor industry for thickness measurement, positioning for exposure, positioning for hard disk assembly

-Turning machine (lathe): measuring rotational motion roughness and surface smoothness of products

-Etc.

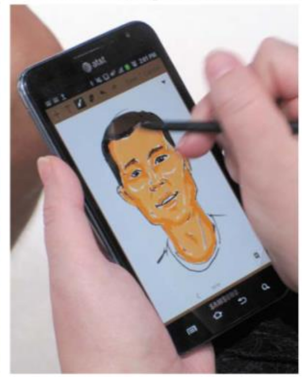
Pros: high accuracy (thickness measurement on <nm scale), can measure in non-contact mode

Cons: requires a clean environment compared to LVDT.

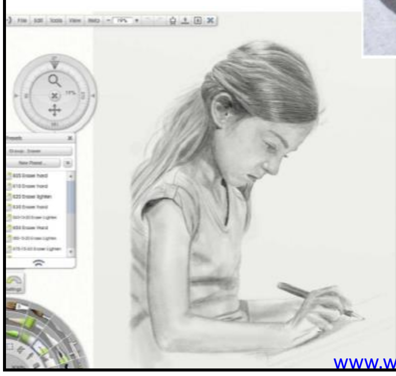
Thin film thickness measurement with nm accuracy, e.g. Alpha-Step IQ Surface Profiler

Touch screen with pressure sensitive pen

Samsung Galaxy Note sketching demo at CES 2012



The Galaxy Notes use both a p-cap touchscreen AND a Wacom EMR stylus (2 sensors!)



Created with an N-Trig active stylus on a Fujitsu Lifebook using ArtRage software

www.walkermobile.com/Touch_Technologies_Tutorial_Latest_Version.pdf (2014)

Capacitive touch screens

Touch screen based on self-capacity

-1 armed electrode with capacitance to ground

-Human body capacity increases electrode capacity to ground

-Each electrode is measured separately

-Applied in cheaper devices

Touch screen based on mutual capacity

-Capacitance between two electrodes

-Human body draws electric field lines, so capacitance decreases

-Each electrode pair crossing is measured separately

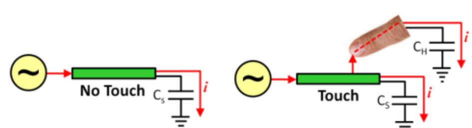
-Position accuracy of 0.1-0.2mm can be achieved

-Transparent electrodes are required for touch screens this makes them relatively expensive.

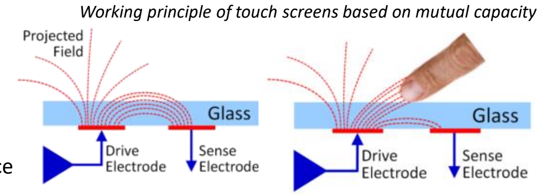
Further sensors for handwriting on a touch screen:

Electromagnetic resonance (EMR) sensor from Wacom. The pressure on the pen is measured by capacitance variation, 1024 different pressure levels are resolved (see next slide).

Operating principle of touch screens based on self-capacitance

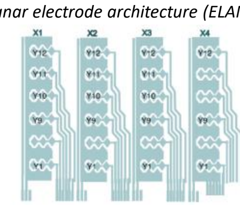


Working principle of touch screens based on mutual capacity



Geometry of a caterpillar-shaped mutual capacitance planar electrode architecture (ELAN)

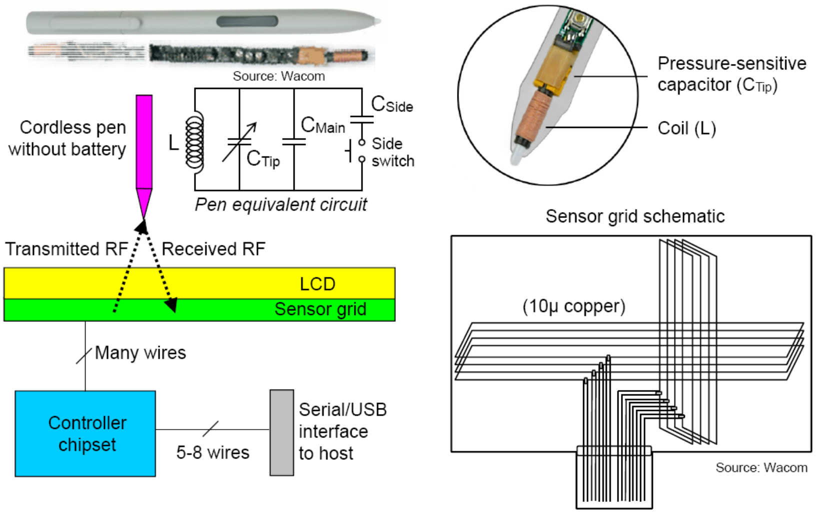
Source: Cypress



www.walkermobile.com/Touch_Technologies_Tutorial_Latest_Version.pdf (2014)

Pressure sensitive touch screen pen

www.walkermobile.com/Touch_Technologies_Tutorial_Latest_Version.pdf (2014)



Under the tablet's surface is a printed circuit board with a grid of multiple send/receive coils. In send mode, the tablet generates a electromagnetic field (~531 kHz), which stimulates oscillation in the pen's coil/capacitor (LC) circuit when brought into range of the field. In receive mode, the resonant circuit's oscillations in the pen is detected by the tablet's grid. This information is analyzed by the computer to determine the pen's position, even if the pen does not touch. In addition, the pressure on the pen detunes the resonance frequency of the LC circuit (and thereby the phase of the response), this is the basis of pressure detection. The tablet provides power to the pen through resonant inductive coupling, no battery is required. (Source of text: Wikipedia)

Laser rangefinder

Amann et al.: Laser ranging: a critical review of usual techniques Optical Engineering, 40 (2001)

Measuring distance with light:

- Interferometric methods (see slides on Metrology): high accuracy
- Time-of-flight (TOF) measurements

Laser rangefinder (LRF) use the TOF method:

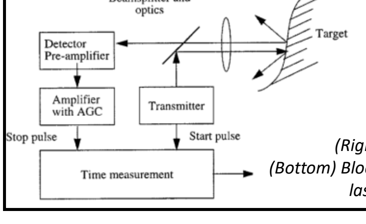
Principle: A laser pulse is sent towards the object, and the time (t_d) it takes for the reflected light to arrive back is measured, $2d = c/t_d$, where d is the distance to the object and c is the speed of light.

Resolution, range: <1mm resolution can be achieved, for this 7ps temporal resolution is required. Can be used at distances of up to several km (e.g. military targets).

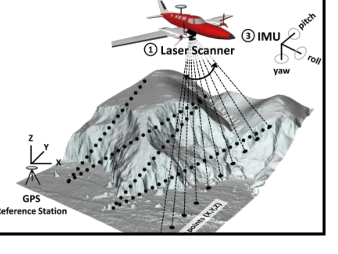
Components: Source: 5-50ns laser pulse, Sensor e.g. avalanche photodiode with automatic gain control (AGC). Time measurement unit: triggered by laser pulse and stopped by incoming signal.

Phase modulation and heterodyne techniques are also applied for laser-based distance measurement (see lab exercise in next semester).

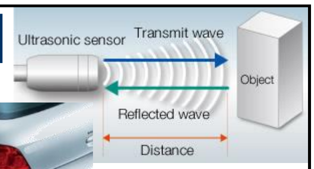
LIDAR (light-radar)
LRF+rotating mirror



(Right) Applications
(Bottom) Block diagram of a laser range finder



Ultrasonic distance detectors and complex object sensing



Ultrasonic distance detectors

- They work on the principle of sonar: they scan the echo of ultrasound.
- Time-of-flight is the measurement principle here too.
- Source not audible range e.g. 18kHz. A simple microphone is the detector.
- Use in cars : detection of short distances, slow motion v_{sound}=350m/s d=30cm → approx. 1ms

Complex distance and environment sensing:

E.g. self-driving cars:

Hhuman error eliminated, 2-3 capacity increase on roads

Car has to deal with a variety of situations → many sensors, e.g.: parking, accident prevention, distance tracking, lane departure monitoring, blind spot monitoring, cross traffic

Distance sensors: -radar (with radio wave, Doppler shift, fast, good for longer distances, weather independent e.g. for adaptive cruising), -ultrasonic sensor, with camera recognition -LIDAR system (light-radar): LRF+rotating mirror (laser, e.g. VelodyneHDL 64 rotating lasers, 1.3 millio points/sec → virtual model of the environment).

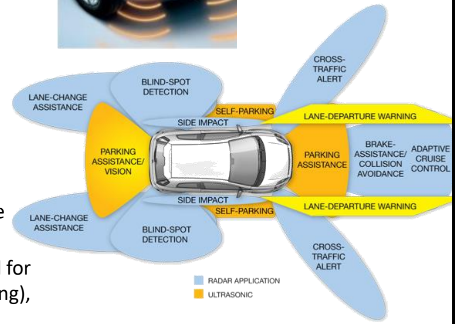
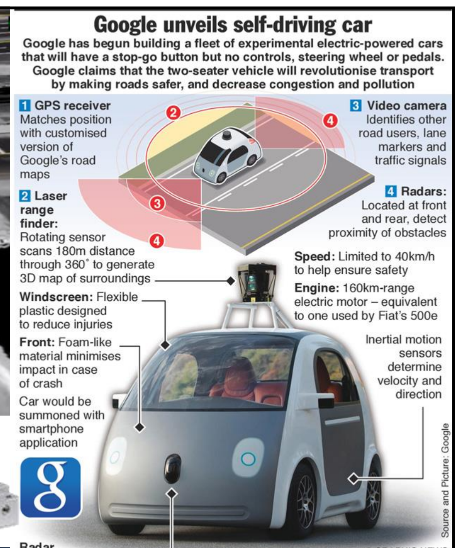


Figure 2 Several driver-assistance systems are currently using radar technology to provide blind-spo detection, parking assistance, collision avoidance, and other driver aids (courtesy Analog Devices).

http://www.edn.com/design/automotive/4368069/Automobile-sensors-may-usher-in-self-driving-cars



...

http://www.edn.com/design/automotive/4368069/Automobile-sensors-may-usher-in-self-driving-cars

Temperature sensors

Primary thermometers: Temperature derived without unknown quantities. E.g.:

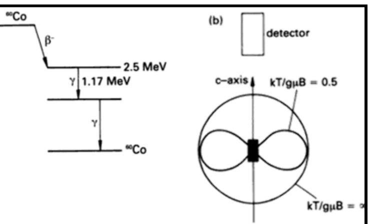
- From thermodynamic properties, such as **sound velocity in noble gases** (see metrology slides)
- **Johnson-Nyquist noise** (see metrology slides)
- **Anisotropic angular distribution of gamma radiation** for a radioactive isotope in B space. Principle: the magnetic moment of a ^{60}Co atom is polarized in B field. In a ferromagnetic ^{59}Co background, the field felt by ^{60}Co is B=10T . Its nuclear spin undergoes hyperfine splitting to N different levels, ΔE~NB, where $\Delta E/k_B$ = 31mK → thus in the mK range the probability of filling different levels is strongly T dependent. At low T ($k_BT<<\Delta E$) only the lowest level is occupied whose gamma decay has a temperature dependent anisotropic angular distribution that can be measured. Sensitivity: good for the mK range, typical calibration procedure for dilution refrigerators.

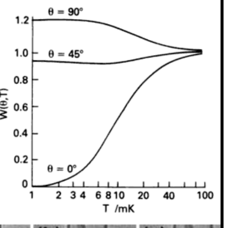
Secondary thermometers: A change in some property of a sensor that is being measured (e.g. resistance, pressure, capacitance). Calibration is required. They usually give a more convenient, more accurately measured signal.

Types: -Thermal expansion based

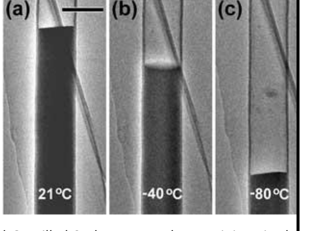
-Thermocouples

-Resistance thermometers , Resistance temperature detectors (RTD), Thermistors,…





(top left) decay process of ^{60}Co, (top right) angular distribution of the gamma decay with respect to the B field direction (bottom) radiation intensity as a function of angle and temperature



A Liquid-Ga-Filled Carbon Nanotube: A Miniaturized Temperature Sensor, Small, 1, 11, 1088 (2005).

http://en.wikipedia.org/wiki/Resistance_thermometer,
http://www.elprocus.com/temperature-sensors-types-working-operation/

Temperature sensors: thermocouples

Thermocouples

Principle: When two conductors of different materials come into contact, a so-called contact voltage is generated. In a chain of conductors using different materials, these contact voltages are cancelled if (i) the two ends are made of the same material (e.g. copper) and (ii) all connection points have the same temperature. However, the contact voltage is temperature-dependent, i.e. a net thermoelectric voltage is measured if one connection point is at the temperature under test (T_{sense}) and the other connection points are at a well-defined reference temperature (T_{ref}). The measured thermoelectric voltage is proportional to the temperature difference, V=S* (T_{sense} -T_{ref}). The phenomenon is based on the Seebeck effect (1821), the sensitivity being determined by the Seebeck coefficient S.

Advantages: cheap, no control voltage required, can be used up to very high temperatures (~2300°C), where components of other, more complex thermometers would be damaged.

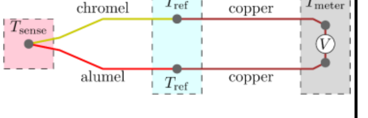
Disadvantages: inaccurate, (≈1C°) the sensitivity is small (<80 µV/°C, see characteristics in the right)

Example: type E thermocouple (68 µV/°C) using Nickel and Chromium-Constantan,

Selection criteria for the thermocouple wires: sensitivity, price, melting point, chemical reactivity

Application: flame sensors, oven thermometers, engine thermometers

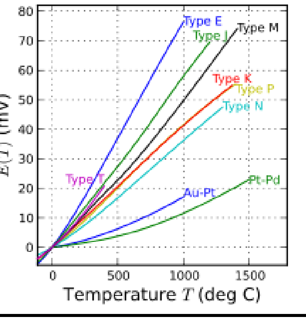
Operation principle of a so-called K-type thermocouple using chromel and alumel wires, i.e. two different alloys the first containing chromium and the last aluminum. This thermocouple has a 41µV/K sensitivity.



$$\nabla V = -S(T)\nabla T,$$

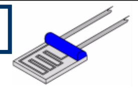
Typical thermocouple thermometer

Response of thermocouples as a function of ΔT



http://en.wikipedia.org/wiki/Thermocouple

Temperature sensors: metallic resistance temperature detectors (RTD)



Resistance thermometer: the thin film Pt wire makes many turns on the sensor surface to increase its resistance

Principle: The temperature is derived by measuring the change in resistance of the sensor (usually metal). Most widely used temperature sensor.

Advantage: Simple principle, easy to measure, low drift, good stability.

Disadvantage: low sensitivity (e.g. $\Delta R/R/\Delta T \sim 0.4\%/C°$ for Pt), slow response.

Metals used: Pt (stable resistance-temp. relationship over a wide T range), Ni (non-linear up to about 300 C°), Cu (most linear, but can oxidize above 150 C°)

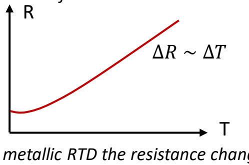
It reproduces so well that it is also used as a temperature measurement standard. See SPRT (standard platinum resistance thermometers) ITS-90.

To obtain a wll-measurable resistance:

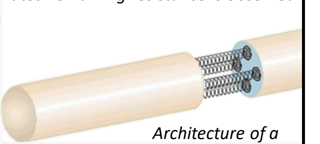
Thin film (1-10nm thick). Thermal expansion difference between substrate/metal results in stress → often a coiled wire is used in a tube filled with gas or ceramic powder to ensure free movement in case of thermal expansion. ITS-90 is also based on this. E.g. in ITS-90 (International Temperature Scales 1990) the Pt SPTR is recommended to be used over a wide T range, see Wiki table.

Sensitivity for a Pt100 sensor: 0.385 Ω/°C. To measure such a small resistance variation, a four probe measurement method is required (see the slides of the first lecture)

What exactly are we measuring the temperature of? E.g. circuit vs. environmental temperature in a mobile phone, or lattice vs. electron temperature in cryogenic meas.



In a metallic RTD the resistance change is proportional to the temperature change over a broad temperature interval, except for the very low temperatures, where a saturated remaining resistance is observed



Architecture of a temperature sensor based on coiled Pt wire

Standard interpolating thermometers and their ranges [edit]

Lower (K)	Upper (K)	Variations	Thermometer	Calibration and interpolation strategy
0.65	3.2	1	Helium-3 vapor pressure thermometer	Vapor pressure–temperature relationship fixed by a specified function.
1.25	2.1768	1	Helium-4 vapor pressure thermometer	Vapor pressure–temperature relationship fixed by a specified function.
13.8033	1234.93	11	Platinum resistance thermometer	Resistance calibrated at various fixed points and interpolated in a specified way. Eleven distinct calibration procedures are specified.

http://en.wikipedia.org/wiki/Resistance_thermometer

Temperature sensors: semiconductor thermometers

Thermistor

It works in a similar way to RTDs but measures semiconductor resistance, which is highly non-linear (see figure). High sensitivity over a small range.

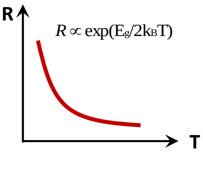
Applications: overheating inhibitors, working-point applications, where a small temp. range should be investigated with high sensitivity, temp. measurement at very low temperatures, where metallic RTDs are already saturated, …

Semiconductor IC thermometers

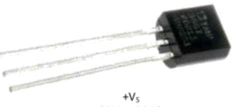
Advantage: sensor is integrated to the circuit, includes ADC, programmable, sufficiently large output signal, …

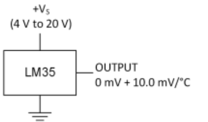
E.g. Texas Instruments LM35 - 1.5$/piece

R – T characteristic of a semiconductor thermometer



Texas Instruments LM35 IC thermometer





Light sensors

Light is a small segment of EM waves: a range that can be perceived by the eye.
$c = f\lambda$, λ= 400-700nm range

Particle nature of light (Einstein 1905, photoelectric effect)
Photon: light transports energy in quanta of E=hf, where h is the Planck's constant.

Wave nature of light: it behaves as a wave according to Maxwell's equations

The particle – wave duality of light is exploited by sensors, some of which are based on the particle nature (photodiodes); some exploit the wave nature (bolometers).

Photodiode

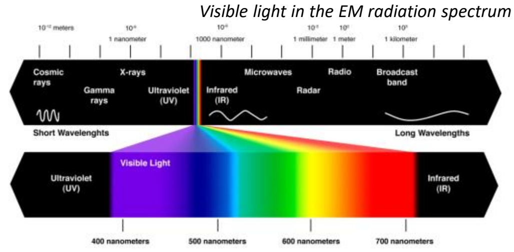
Semiconductor diode in PN or PIN junction (I insulator) with reverse bias

Principle (photoelectric effect): iIf the energy of the incoming photon is greater than the band gap (i.e. E>Eg), it excites electrons from the valence band band into the conduction band. The applied voltage V causes the electrons to leave the electrons towards the N-doped cathode side and the holes towards the P-doped anode, generating a photocurrent.

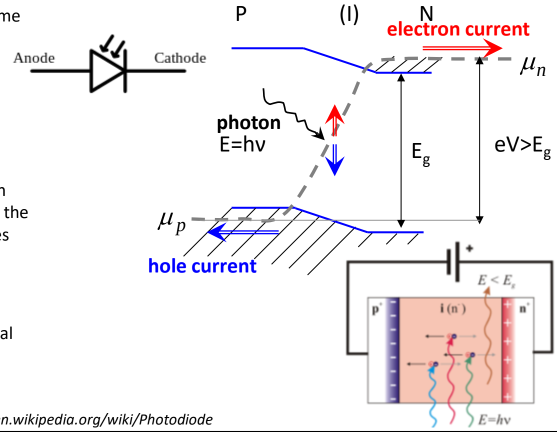
Biasing increases response speed (RC term) but also noise background (dark current).

Applications: CD readers, remote controls, optocouplers, optical communication, ...

Visible light in the EM radiation spectrum



Photodiode circuit diagram and operating principle



Blackburn, Modern Instrumentation for Scientists and Engineers, http://en.wikipedia.org/wiki/Photodiode

Light sensors

CCD (Charge Coupled Device)

1969 Bell Labs, Boyle and Smith (2009 Nobel prize)

Principle: Light creates trapped electrons in a MOS capacitor array, where each capacitor is a pixel of the sensor. Then, using the pixels as a shift register, the trapped electron can be delivered to the edge of the sensor, where a charge amplifier can be used to convert each pixel signal into a voltage signal and store it one by one.

CMOS Active Pixel sensors

Each pixel contains a photodiode (PPD, pinned photodiode, p+/n/p junctions) and an active amplifier.

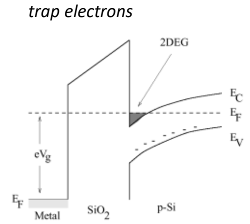
Mobilephone cameras, webcams and digital cameras after 2010 use this technology.

Advantages: cheaper, lower power consumption than CCD, easier fabrication, faster readout, can do both sensing and processing, less crosstalk to neighbouring pixels than in CCD.

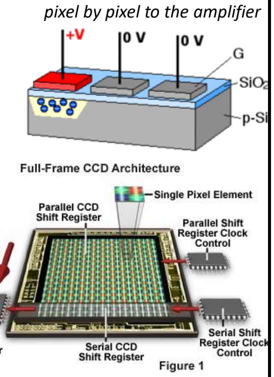
Further light sensors:

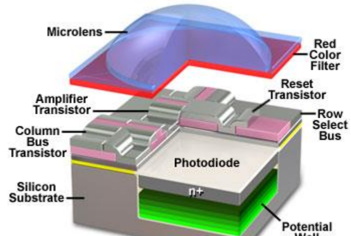
Photoresistor, avalanche photodiode single photon detector, photomultiplier tube, ...

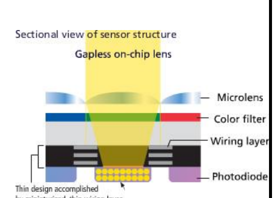
MOS (Metal /oxide / semiconductor) band structure with inversion layer on the surface. The potential well can trap electrons



Charge coupled principle, as the trapped charges are delivered pixel by pixel to the amplifier







Nikon CMOS sensor (2012) pixel size <6um

Sensing electromagnetic radiation: bolometers

Principle: It contains an absorption element (e.g. a thin layer of metal) which, when exposed to radiation, increases the temperature which is measured. It is sensitive to the energy left in the detector. The detector is usually connected to a cooling tank.

1880 Langley first bolometer. Platinum band, Wheatstone bridge arrangement. He could detect 1 cow at half a mile.

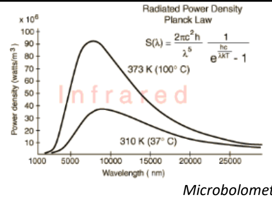
Used in the infrared and terahertz range (λ= µm - mm).

Application: astronomy, particle physics (cooled detectors), Thermal imaging cameras, ...

Mikrobolometer (in thermal imaging cameras):

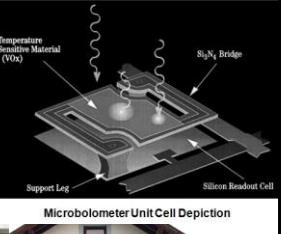
The sensor is built with resistance measurement electronics integrated in a silicon chip and a suspended vanadium oxide or amorphous silicon wafer above (~1µm), which varies its resistance due to the irradiation. Suspension provides thermal insulation from the chip. Sometimes a heat mirror is put on the Si substrate (e.g. titanium) under the wafer. Typical resolution: 1024x768

Uses: thermal engineering, medical diagnostics, military, security engineering, automotive (animal detection), ...

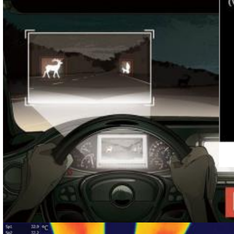


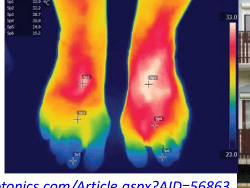
Black body radiation at different temperatures. At 37 C° maximum intensity around 10µm wavelength.

Microbolometer design: suspended VOx plate above the chip



Autoliv: night time wildlife detection, signals and spot light illumination. Mercedes S-Class (2014)





(left) Thermal camera at airport to detect fever from H1N1 virus,
(right) Diagnosing inflammation due to increased blood flow
(right 2) Thermal camera image of apartment building in winter.

http://www.photonics.com/Article.aspx?AID=56863,
http://en.wikipedia.org/wiki/Microbolometer

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Microelectromechanical systems (MEMS) as acceleration sensors

Acceleration sensor:

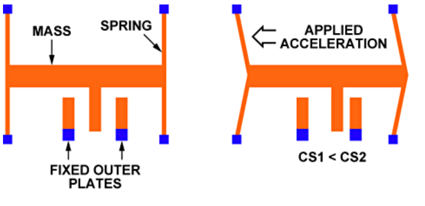
Principle: A fixed plate provides one electrode of a capacitor, the other electrode is suspended on micromachined springs, having a well-defined mass. When the sensor accelerates in x direction, the mass is displaced by the amount determined by the springs. Due to the displacement, the capacitance between the two electrodes. On the figure two fixed electrodes are used, CS1 capacitance decreases and C2 increases due to the displacement.

Silicon-based devices (good mechanical properties, rigid), can be made using IC manufacturing techniques. Suspended structures are created over a silicon chip (using a sacrificial layer)

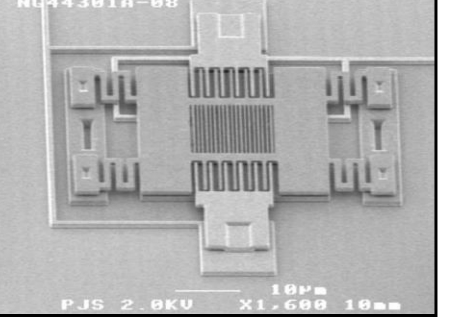
Pl. ADXL321 4mmx4mm, resolution: 3mg@50Hz, range: ±18g, X &Y axis

Static acceleration ("g") and dynamic acceleration are both measured. Price: 8$.

Operation of MEMS accelerometers: Capacitive measurement of the displacement of a mass on a spring



SEM image of accelerometer with mass suspended on springs in the middle, and capacitive readout at the sides.



Blackburn, Modern Instrumentation for Scientists and Engineers,
http://www.memsjournal.com/2011/01/motion-sensing-in-the-iphone-4-mems-gyroscope.html
http://www.digikey.com/en/articles/techzone/2011/apr/mems-accelerometers-gyroscopes-and-geomagnetic-sensors---propelling-disruptive-consumer-applications

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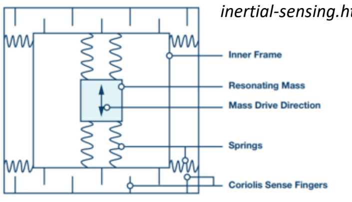
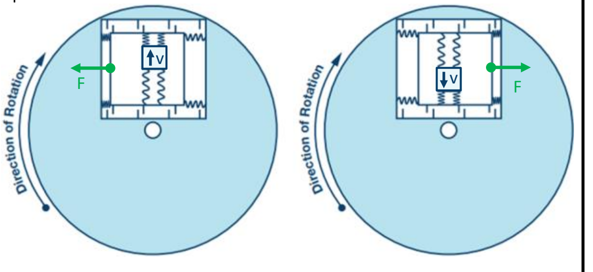
MEMS gyroscopes

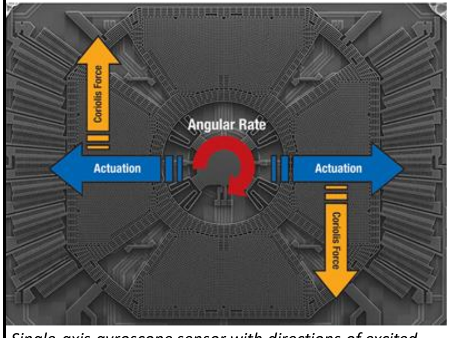
Source of text and illustrations: https://www.analog.com/en/resources/technical-articles/mems-gyroscope-provides-precision-inertial-sensing.html

Principle: The Coriolis force on a resonating mass is proportional to the angular velocity ($\boldsymbol{F}_C = -2m\boldsymbol{\Omega} \times \boldsymbol{v}$).
When the resonating mass moves toward the outer edge of the rotation, it is accelerated to the right and exerts on the frame a reaction force to the left. When it moves toward the center of the rotation it exerts a force to the right, as indicated by the green arrows. The frame containing the resonating mass is tethered to the substrate by springs at 90° relative to the resonating motion, as shown in the Figure. This figure also shows the Coriolis sense fingers that are used to sense displacement of the frame through capacitive transduction in response to the force exerted by the mass.

E.g. ST L3G4200D, 3axis digital output, size: 4x4x1mm, Sensitivity 8m°/sec, max range 2000 °/sec







Single-axis gyroscope sensor with directions of excited vibration, rotation and Coriolis force, STMicroelectronics

Blackburn, Modern Instrumentation for Scientists and Engineers, http://www.memsjournal.com/2011/01/motion-sensing-in-the-iphone-4-mems-gyroscope.html http://www.digikey.com/en/articles/techzone/2011/apr/mems-accelerometers-gyroscopes-and-geomagnetic-sensors---propelling-disruptive-consumer-applications

MEMS acceleration sensors and gyroscopes

STMicoelectronics multi-axis gyroscope

Usage:

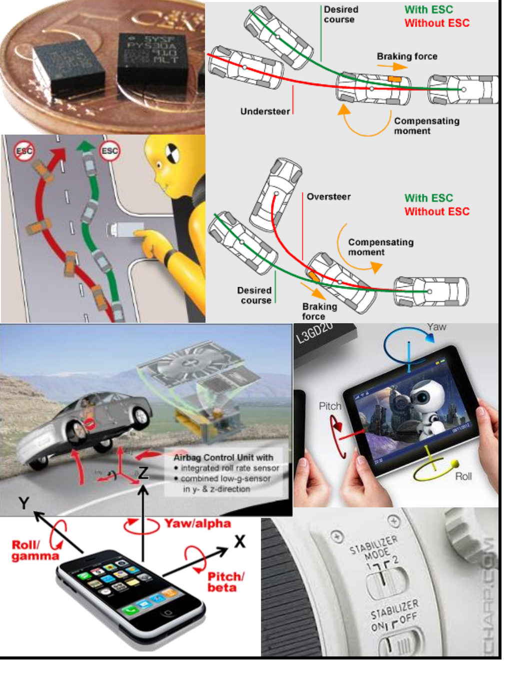
Engine acceleration or vibration, airbag controls (collision, upside down), corner stabilizers, ESC (electronic stability control), braking system activation in case of sudden change of direction, Automatic collision notification (Toyota, Lexus, BMW, Ford, ...)

Earthquake forecasting, building stability testing, navigation systems (in the absence of GPS)

Orientation of mobile devices (standing/lying), FFS (free fall sensor) in notebook hard disks (Dell, Lenovo, Apple, WD ...) - picks up the reading head from the disk. Nintendo playstation - hand movement tracking.

Image stabilisation for cameras

Gravitational field meters



Piezoelectric acceleration sensors

Piezoelectricity: Mechanical stress → deformation → surface charge appears on piezoelectric material.

E.g. in the case of quartz (SiO_2), due to the crystalline arrangement of the atoms as shown in the figure (see quartz clocks, ultrasonic generators, etc.)

Acceleration can be converted to force measurement. At constant force, the charge leaks (leakage resistance), so it is not suitable for measuring dc force. In ac, it can operate up to high frequencies.

Types:

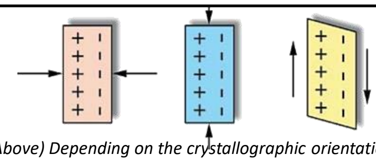
a) Fixed mass on a piezo block

b) Shear piezo, mass is on the side of the piezo.

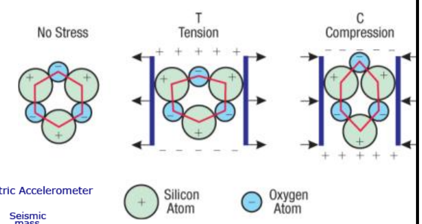
Typical parameters:

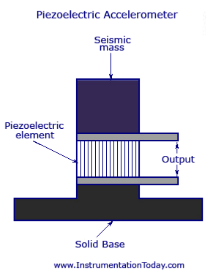
Range can reach 10000g, 1Hz - >10kHz, 10-100mV/g

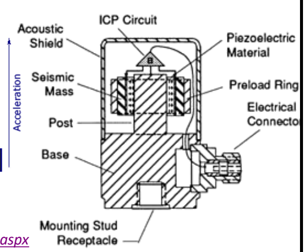
Use: in shock and collision sensors



(Above) Depending on the crystallographic orientation of the quartz, different piezoelectric phenomena can be observed (A) longitudinal, (B) transverse, (C) shear. (Below) Microscopic origin of the transverse case.







Accelerometers structure: inertial mass attached to the piezoelectric material with longitudinal piezoelectric effect on the left and shear piezoelectric effect on the right.

Blackburn, Modern Instrumentation for Scientists and Engineers, https://www.pcb.com/TechSupport/Tech_Accel.aspx
http://www.sensorsmag.com/sensors/acceleration-vibration/the-principles-piezoelectric-accelerometers-1022